

ICT305 Data Visualisation and Simulation

Do Countries That Contribute Less to Climate Change Suffer More From Its Impacts?

Exploring the unequal burden of climate change across countries

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Table of Contents

1. Introduction	3
2. Dataset Description and Justification	3
2.1.Dataset Overview	3
2.2.Dataset Justification	4
2.3.Interactivity	4
3. Stakeholders and Decision Context	4
4. Dashboard Design and Structure	5
4.1.Dashboard Layout and Navigation	5
4.2.Dashboard Overview	5
4.3.Dashboard Sections and Visualisations	5
4.4.Design Rationale	7
5. Key Insights	8
6. Recommendations	9
7. Technical Implementation	11
8. Limitations	11
9. Conclusion	12
10. References	13

1. Introduction

Climate change is widely recognised as a global issue; however, its impacts are not distributed equally across countries. Many lower-emitting nations face greater environmental exposure, population displacement, and adaptation challenges despite contributing less to global carbon emissions.

This project investigates climate inequality through the development of a Python-based interactive dashboard that supports exploration of climate vulnerability, population exposure, and economic adaptation capacity. The dashboard enables users to explore relationships between climate exposure, economic readiness, and adaptation needs through interactive visualisations and filtering.

The central guiding question of this dashboard is:

Do countries that contribute less to climate change suffer more from its impacts?

2. Dataset Description and Justification

2.1.Dataset Overview

This dashboard combines multiple real-world datasets to investigate climate exposure and adaptation readiness across countries.

Datasets:

- World Bank Open Data - Population Dataset
- World Bank Climate Change Knowledge Portal - Population Exposure and Density Dataset
- NASA Sea Level Projection Tool
- World Development Indicators - GDP per Capita
- Our World in Data - Annual CO₂ Emissions

Variables incorporated include:

Environmental Variables

- Sea level change
- Climate projection scenarios
- Population exposure

Socioeconomic Variables

- GDP per capita
- Population density
- Economic adaptation capacity

Climate Responsibility Variables

- Annual CO₂ emissions

2.2.Dataset Justification

Climate change is a multidimensional problem that cannot be explained using a single indicator. The selected datasets combine environmental, demographic, socioeconomic, and emissions data to provide a more comprehensive view of climate vulnerability. Integrating these variables allows the dashboard to examine relationships between climate responsibility, population exposure, and adaptation capacity rather than analysing each factor independently.

Static visualisations cannot adequately reveal these relationships because stakeholders need to compare countries, filter vulnerability categories, evaluate future climate scenarios, and investigate trade-offs between exposure and adaptive capacity. Interactive analysis supports exploratory decision-making by allowing users to investigate multiple perspectives rather than relying on fixed conclusions.

2.3.Interactivity

Interactivity was essential because stakeholders need to:

- Adjust country rankings
- Compare multiple sea-level scenarios
- Explore different vulnerability categories
- Dynamically investigate country performance

Without interaction, users would be limited to fixed conclusions.

3. Stakeholders and Decision Context

Intended Stakeholders

Governments

- Use insights to prioritise adaptation investment and relocation planning.

International Climate Agencies

- Allocate climate support funding.

NGOs

- Identify vulnerable populations requiring assistance.

Policymakers

- Develop climate resilience policies.

Decision Context

This dashboard primarily supports Strategic Decision Making by allowing:

- Long-term climate adaptation planning
- Infrastructure investment
- Allocation of climate funding
- Population protection strategies

The dashboard supports evidence-based strategic planning by enabling stakeholders to compare climate exposure, adaptive capacity, and future climate scenarios before allocating resources.

4. Dashboard Design and Structure

The overall design follows the principles of clarity, consistency, accessibility, and effective information visualisation. Each visualisation addresses a specific analytical question while maintaining a consistent layout, colour scheme, and interaction style throughout the dashboard. This ensures users can easily interpret the visualisations without repeatedly learning new interfaces or legends.

4.1. Dashboard Layout and Navigation

The dashboard follows Shneiderman's visual information seeking mantra of "overview first, zoom and filter, then details on demand." Users begin with a dashboard overview before progressively exploring individual analytical views through tab navigation. Sidebar filters provide dynamic interaction without overwhelming the interface, enabling users to customise the analysis while maintaining a consistent workflow. This structure reduces cognitive load and supports efficient exploratory analysis.

Control	Function	Purpose
Ranking slider	Adjusts the number of countries shown in ranking charts	Allows users to compare a smaller or larger group of countries
Risk zone filter	Filters countries by vulnerability category	Helps users focus on specific risk groups such as Critical Risk or Lower Risk
Sea-level scenario selector	Allows users to select climate projection pathways	Supports comparison of different future climate outcomes
Country selector	Filters the country-level data table	Allows users to inspect specific countries in detail

4.2. Dashboard Overview

The dashboard begins with an overview section that gives users a quick summary of the dataset and key risk indicators. This section includes four key performance indicators:

- Countries analysed
- Critical risk countries
- Highest pressure country
- Average population exposed

These indicators provide an overview of the dataset and establish context before users begin detailed exploration.

4.3. Dashboard Sections and Visualisations

The dashboard contains five main analytical views and one supporting data table. Each section contributes to the overall research question: Do countries that contribute less to climate change suffer more from its impacts?

Section	Visualisation	Purpose
1	Population Exposure Ranking	Identifies countries with the highest displacement pressure
2	Global Population Density Map	Shows where population concentration is geographically located
3	Sea Level Trend	Compares historical and projected sea-level change under different scenarios
4	Adaptation Capacity Scatter Plot	Compares population exposure, GDP per capita, and annual CO ₂ emissions
5	Risk Matrix	Identifies countries most in need of adaptation support
6	Country-level Data Table	Provides access to the underlying indicators used in the dashboard

Visualisation Selection Justification

Each visualisation was selected to communicate a different aspect of climate vulnerability. Ranking, geographic, temporal, and multivariate visualisations are combined to provide complementary perspectives, while the interactive data table improves transparency by allowing users to inspect the underlying indicators.

Visualisation 1: Population Exposure Ranking

The horizontal bar chart was selected for the Population Exposure Ranking because it provides the clearest comparison of country rankings by displacement pressure. Displacement pressure is calculated using:

Displacement Pressure Index = $\log(1 + \text{Population Exposed}) \times \log(1 + \text{Population Density})$

The Displacement Pressure Index combines population exposure and population density using logarithmic transformation to reduce the influence of extreme values while preserving meaningful differences between countries. Countries are coloured by risk zone, allowing users to identify which countries face both high exposure and weaker economic capacity.

Visualisation 2: Global Population Density Map

The choropleth map showing global population density was chosen to communicate the geographic distribution of population density, enabling users to identify regional patterns that cannot easily be observed through tabular data. This map helps users understand where displacement pressure is geographically concentrated. Population density is included because densely populated regions have less capacity for relocation and experience greater disruption when climate hazards occur.

Visualisation 3: Sea Level Trend

The line chart is appropriate for displaying historical and projected sea-level trends because it clearly communicates historical and projected global mean sea-level change under multiple climate scenarios. It compares historical observations with future projection scenarios, including strong climate action, moderate emissions, and very high emissions

pathways. This chart helps users understand that displacement pressure is not only a current issue but also a future risk. The widening gap between scenarios demonstrates that climate decisions made today can influence the severity of future impacts.

Visualisation 4: Adaptation Capacity

This chart compares climate exposure, economic capacity, and emissions to reveal whether highly exposed countries also possess the resources needed to adapt. As this chart has multiple factors, a Scatter Plot is chosen to represent the data. GDP per capita is used as a proxy for economic adaptation capacity, while bubble size represents annual CO₂ emissions.

Visualisation 5: Risk Matrix

Combines displacement pressure and economic capacity into a vulnerability matrix. Similarly to the Adaptation Capacity, multiple data points are represented hence the usage of a Scatter plot. Countries are positioned based on displacement pressure and GDP-based capacity, allowing users to identify four broad risk groups:

- **Critical Risk:** High displacement pressure and lower economic capacity
- **High Exposure, Higher Capacity:** High displacement pressure but stronger economic capacity
- **Emerging Risk:** Lower economic capacity but currently lower displacement pressure
- **Lower Risk:** Lower displacement pressure and stronger economic capacity

This matrix is the main decision-support view because it identifies where adaptation support may be most urgently needed.

Supporting Data Table

A country-level data table improves transparency by allowing users to inspect the indicators behind the visualisations, including population exposure, population density, displacement pressure, GDP per capita, annual CO₂ emissions, and risk category. The table also supports further analysis by allowing users to download the filtered dataset.

4.4.Design Rationale

Several design principles were applied consistently throughout the dashboard to improve usability and support effective decision-making.

Clarity

Each visualisation addresses a single analytical question and is accompanied by a concise explanation describing its purpose. Insight panels beneath every chart summarise the key findings and reduce the need for users to interpret complex graphics independently.

Consistency

A consistent visual language is maintained throughout the dashboard. Risk categories use identical colours across every visualisation, while fonts, layouts, spacing, and chart styles remain uniform. This consistency improves recognition and reduces cognitive effort when navigating between sections.

Accessibility

The dashboard was designed for both technical and non-technical audiences. Plain language headings, descriptive subtitles, interactive tooltips, and explanatory panels improve readability, while sidebar filters allow users to focus on information relevant to their needs.

Information Storytelling

The dashboard follows a structured storytelling approach that guides users from identifying where climate vulnerability exists, to understanding why it occurs, how it may change in the future, and which countries require the greatest support. This progression transforms individual visualisations into an integrated analytical narrative that supports informed decision-making.

5. Key Insights

Collectively, the dashboard reveals three key findings. First, climate impacts are distributed unequally across countries. Second, vulnerability is shaped by both environmental exposure and economic capacity rather than emissions alone. Third, future climate risks depend heavily on present-day mitigation efforts, highlighting the importance of balancing emissions reduction with adaptation planning. The following discussion explains how each visualisation contributes to these findings.

Climate impacts are distributed unequally

The Population Exposure Ranking demonstrates that countries experiencing the greatest displacement pressure are not necessarily those producing the highest greenhouse gas emissions. Countries such as Bangladesh, India, Pakistan, and Sri Lanka consistently appear among the highest-ranked nations because they combine large exposed populations with extremely high population densities.

This finding highlights that climate vulnerability depends on exposure rather than emissions alone. While many developed nations contribute substantially more to cumulative emissions, several developing countries face greater immediate risks because environmental hazards affect far larger populations living within limited geographic space.

Population concentration amplifies displacement risk

The Global Population Density Map illustrates that displacement pressure is geographically concentrated rather than evenly distributed around the world.

Regions across South Asia, East Asia, and parts of Europe display significantly higher population densities than many other parts of the world. When climate hazards such as sea-level rise, coastal flooding, or storm surges occur, densely populated regions experience greater social disruption because larger numbers of people require evacuation, housing, and infrastructure support simultaneously.

Conversely, countries with lower population densities generally possess greater spatial flexibility for adaptation and relocation. The spatial distribution shown by the map explains why several countries consistently appear among the highest displacement pressure rankings.

Future climate scenarios increase uncertainty

The Sea Level Trend chart demonstrates that all future climate scenarios project continued sea-level rise throughout the coming century. Although strong climate mitigation (SSP1-2.6) substantially slows the rate of increase compared with high-emission pathways (SSP5-8.5), none of the scenarios indicate stabilisation within the projection period.

Importantly, the divergence between scenarios widens considerably after the present day. This suggests that decisions made over the coming decades will significantly influence long-term displacement pressure, adaptation costs, and population exposure. The chart therefore highlights that future climate risk is not fixed but depends heavily upon present-day policy decisions.

Economic capacity strongly influences resilience

The Adaptation Capacity chart compares population exposure against GDP per capita while representing annual carbon emissions through bubble size. The visualisation reveals that countries experiencing similar levels of exposure often possess very different capacities to respond. Higher-income countries generally possess greater financial resources to construct flood protection, strengthen infrastructure, relocate vulnerable communities, and recover from climate-related disasters. In contrast, lower-income countries may struggle to finance adaptation despite facing comparable or greater environmental risks. The chart also demonstrates that several highly exposed countries are relatively small contributors to annual carbon emissions, reinforcing the concept of climate inequality.

Climate responsibility differs from climate vulnerability

The Risk Matrix combines displacement pressure with economic capacity to classify countries into four broad categories. Countries within the Critical Risk quadrant experience both high displacement pressure and relatively lower economic readiness. These countries represent those requiring the greatest priority for international adaptation support. Population exposure, adaptive capacity, and future climate projections combine to create unequal climate impacts, demonstrating that countries most vulnerable to climate change are often not those most responsible for causing it.

6. Recommendations

Stakeholders should interpret the dashboard as a sequential decision-support tool rather than a collection of independent charts. Users should first examine the Population Exposure Ranking to identify countries facing the greatest displacement pressure, then use the Global Population Density Map to understand the geographic distribution of vulnerable populations. The Sea Level Trend provides future climate context by comparing projected outcomes under different emissions scenarios, while the Adaptation Capacity chart enables comparison between environmental exposure and economic readiness. Finally, the Risk Matrix assists stakeholders in prioritising countries requiring immediate adaptation support based on both vulnerability and adaptive capacity. Based on the dashboard findings, the following recommendations are proposed for key stakeholders.

Governments

Governments should prioritise adaptation strategies in regions experiencing the highest displacement pressure. Rather than distributing resources evenly, investment should focus on vulnerable coastal communities where environmental exposure and population density are greatest.

Recommended actions include:

- Expanding coastal defence infrastructure such as seawalls and flood barriers.
- Developing long-term relocation and land-use planning for vulnerable communities.
- Investing in climate-resilient housing, transportation networks, and emergency shelters.
- Improving early warning systems and disaster preparedness programmes.

International Climate Organisations

International organisations such as the United Nations, World Bank, Green Climate Fund, and World Health Organization should allocate climate adaptation funding based on vulnerability rather than emissions alone. Countries classified as Critical Risk require greater financial and technical assistance because they face both high displacement pressure and limited economic capacity.

Priority should therefore be given to:

- Adaptation grants rather than loans.
- Coastal protection projects.
- Community relocation programmes.
- Capacity building for local governments.
- Climate resilience infrastructure.

Non-Government Organisations (NGOs)

NGOs should use the dashboard to identify vulnerable populations and strengthen disaster preparedness before climate-related disasters occur.

Recommended actions include:

- Prioritising disaster preparedness programmes.
- Developing community resilience initiatives.
- Supporting evacuation planning.
- Expanding humanitarian relief capacity in high-risk regions.
- Educating local communities about climate adaptation strategies.

Policymakers

Climate policies should recognise that countries experiencing the greatest climate impacts are not always those contributing the most greenhouse gas emissions. Future policies should therefore balance emissions reduction with adaptation financing to support vulnerable nations.

Policymakers should therefore:

- Increase international adaptation funding for vulnerable developing nations.
- Incorporate climate vulnerability into funding allocation frameworks.
- Encourage technology transfer for climate adaptation.
- Strengthen international cooperation on displacement planning and climate resilience.
- Develop policies that consider both environmental exposure and socioeconomic vulnerability.

Such policies would contribute towards a more equitable global response to climate change.

Researchers and Future Development

Although the dashboard provides valuable insights, future work could improve its analytical capability through additional datasets and predictive modelling.

Potential improvements include:

- Incorporating sub-national and city-level vulnerability data.
- Including governance, healthcare, education, and infrastructure indicators.
- Integrating real-time climate monitoring through APIs.
- Applying predictive analytics or machine learning models to estimate future displacement.
- Developing interactive regional drill-down functionality.

These enhancements would strengthen the dashboard's usefulness for long-term planning and policy development.

7. Technical Implementation

The dashboard was developed using Python, Streamlit, Pandas, NumPy, and Plotly Graph Objects. These technologies provide a reproducible environment for interactive data processing and visualisation. Data preprocessing included cleaning missing values, standardising country names, applying logarithmic transformations, grouping climate projection scenarios, and generating derived indicators such as the Displacement Pressure Index and risk classifications. These steps improved data consistency while reducing the influence of extreme values.

Interactive Features

The dashboard includes several interactive capabilities:

- Ranking sliders
- Multi-select filters
- Climate scenario selectors
- Country filtering
- Interactive Plotly tooltips
- Zooming and panning

These interactions encourage users to perform exploratory analysis rather than simply observing static graphics.

8. Limitations

Although the dashboard provides valuable insights into climate inequality, several limitations should be acknowledged.

Limitation	Explanation
Country-Level Aggregation	The dashboard analyses countries as single units. Sub-national variation within countries cannot be observed, meaning vulnerable coastal communities may be hidden within national averages.
Projection Uncertainty	Future sea-level projections are based on climate models and therefore contain inherent uncertainty. Actual outcomes will depend upon future greenhouse gas emissions, technological development, and international climate policy.

Limitation	Explanation
Data Availability	Several indicators contain missing values for some countries. These observations were removed during preprocessing, reducing the number of countries included within certain visualisations.
Limited Adaptation Indicators	Economic readiness is primarily represented using GDP per capita. Although GDP provides a useful approximation of adaptive capacity, resilience is also influenced by governance quality, healthcare systems, education, infrastructure, and institutional effectiveness. Including these variables could provide a more comprehensive measure of adaptation capacity.

Although the dashboard provides meaningful decision support, it should complement rather than replace expert judgement. Climate vulnerability is influenced by political stability, governance, infrastructure quality, healthcare systems, and local environmental conditions that cannot be fully represented within a country-level analytical dashboard. Decision-makers should therefore interpret dashboard findings alongside regional assessments and local expertise.

Future versions of the dashboard could include:

- Regional drill-down analysis
- Time-series animation
- Machine learning predictions
- Hazard-specific displacement modelling
- Additional climate indicators
- Live API integration for continuously updated datasets

These improvements would further strengthen the dashboard's decision-support capability.

9. Conclusion

Climate change is not only an environmental issue but also a global equity challenge. By integrating population exposure, sea-level rise, economic capacity, and carbon emissions, the dashboard demonstrates that countries experiencing the greatest climate impacts are often not those contributing the most greenhouse gas emissions. Interactive visualisations, consistent design, and modular implementation enable users to explore these relationships while supporting evidence-based adaptation planning. Overall, the dashboard transforms multiple climate datasets into meaningful insights that promote more equitable climate policy and resource allocation.

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